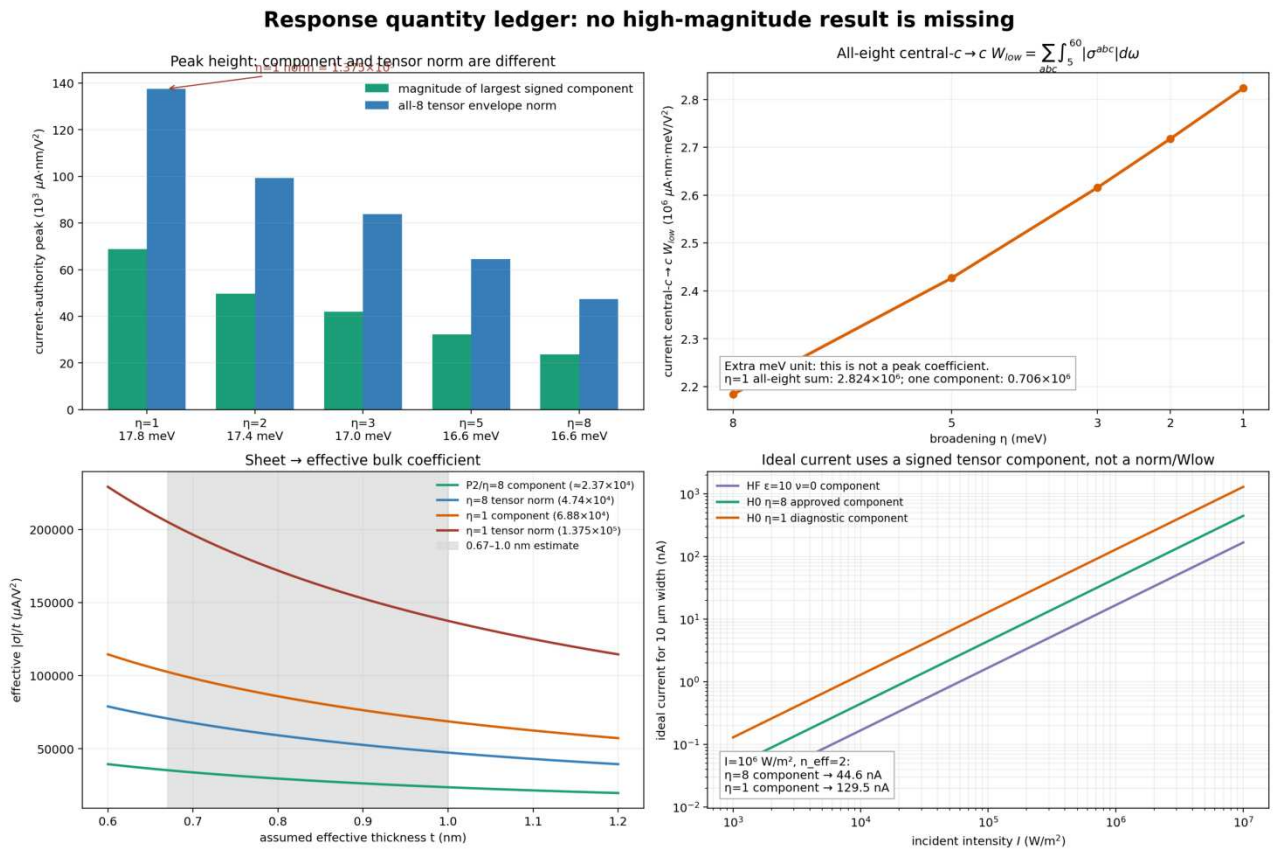


HTQG 巨大体光伏效应

拓扑、序参量与响应量级账本

Revision: component / tensor norm / integrated Wlow separated



- Type-I $\alpha\beta\alpha$ 的错误零频大电流仍明确排除 ($\hbar\omega < 5 \text{ meV}$)。
- Largest clean $\eta=1$ component = 6.876×10^4 ; tensor norm = $1.375 \times 10^5 \mu\text{A}\cdot\text{nm}/\text{V}^2$.
- All-eight central-c $\rightarrow$ c $W_{\text{low}} = 2.824 \times 10^6 \mu\text{A}\cdot\text{nm}\cdot\text{meV}/\text{V}^2$; 不是 peak coefficient。
- P2-approved headline component = $2.3678 \times 10^4 \mu\text{A}\cdot\text{nm}/\text{V}^2$ at 16.6 meV .

先分清三个量，再讨论“最大光电流”

- Single component $\sigma^{\{a;bc\}}(\omega)$: 可在指定 polarization/current axis 下进入 photocurrent。
- Tensor-envelope norm $||\sigma||$: 所有分量的比较标量，不是一项 signed measurable component。
- W_{low} 必须注明 channel/component scope; 多一个 meV, 不是 peak coefficient。

Current-authority $\eta=1$ reconciliation

Largest single component	6.876e4 $\mu\text{A}\cdot\text{nm}/\text{V}^2$
All-eight tensor-envelope norm	1.375e5 $\mu\text{A}\cdot\text{nm}/\text{V}^2$
All-eight central-c $\rightarrow$ c W_{low}	2.824e6 $\mu\text{A}\cdot\text{nm}\cdot\text{meV}/\text{V}^2$
Ideal current must use a signed component, not norm/ W_{low} .	

Why the old graph looked different

- Old $\eta=1$ historical component 1.375e5 $\rightarrow$ current component 6.876e4 after $\cdot 0.5$ 。
- Current tensor norm is 1.375e5 because four large symmetry-related components give norm $\approx 2\times$ one component。
- Old B2 heatmap: one-component W_{low} up to current 0.706e6; all-eight central-c $\rightarrow$ c sum is 2.824e6。

Type-II 比 Type-I 明显更亮；错误零频不再展示

- 左上 x axis 从 5 meV 开始； $\alpha\beta\alpha$ spurious zero-frequency maximum 被明确排除。
- 右上 $\alpha\beta\gamma$ 在 16.6 meV 达到 current envelope norm $4.7397e4 \mu\text{A}\cdot\text{nm}/\text{V}^2$ 。
- Type-II active8/active12 overlap；Type-I remote active8 peak 不通过 window test。

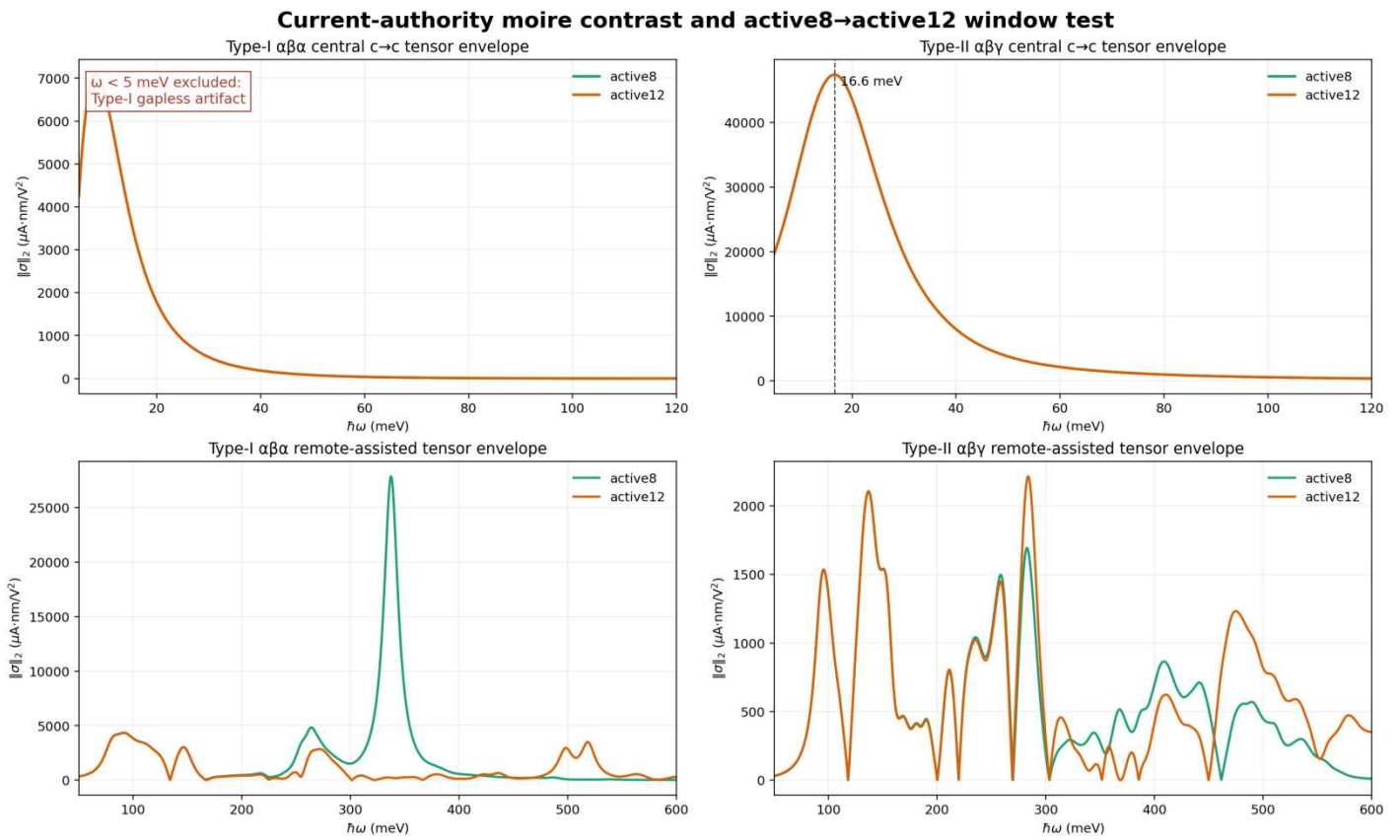


图 1 | Current-authority moire contrast; Type-I $\hbar\omega < 5$ meV excluded.

旧图的 1e5 分量、当前 1e5 norm 与 1e6 Wlow

- Figure 2 historical curves are one dominant component: $\eta=1$ old $1.375e5 \rightarrow$ current $6.876e4$.
- Same $\eta=1$ spectrum: all-eight norm $1.375e5$; all-eight central-c $\rightarrow$ c Wlow $2.824e6$ with extra meV.
- These are not missing results; they are different response summaries and must not be mixed.
- Direct/indirect gaps = $3.4795/1.5630$ meV; candidate_near_degenerate_group=false.

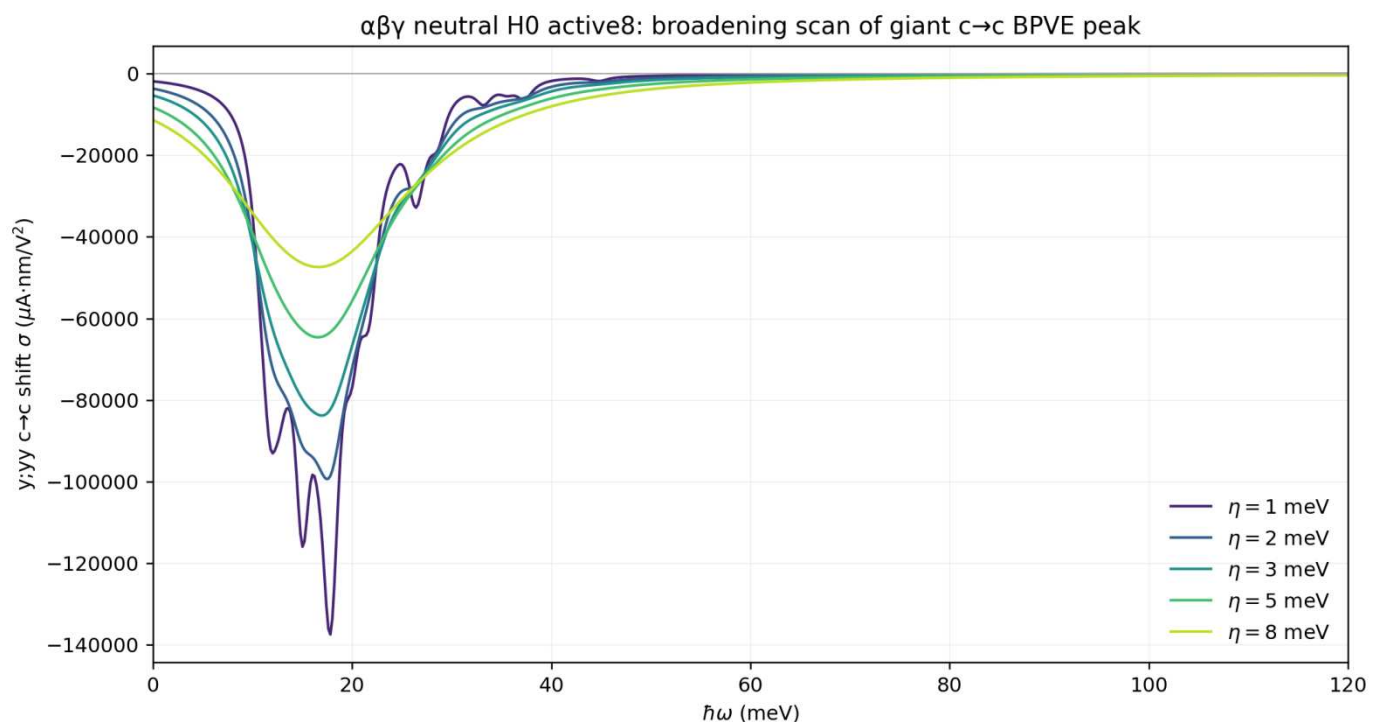


图 2 | Historical one-component broadening scan; current component, norm, and Wlow are tabulated separately.

Historical Figures 2-9: signed $\rightarrow$ -old/2; norms/absolute weights $\rightarrow$ old/2

18×18、24×24、30×30 曲线重合

- Peak position、line shape 与 integrated weight 在 mesh refinement 下稳定。
- 与 finite gaps 和 active8→active12 overlap 一起，排除离散网格偶然热点解释。

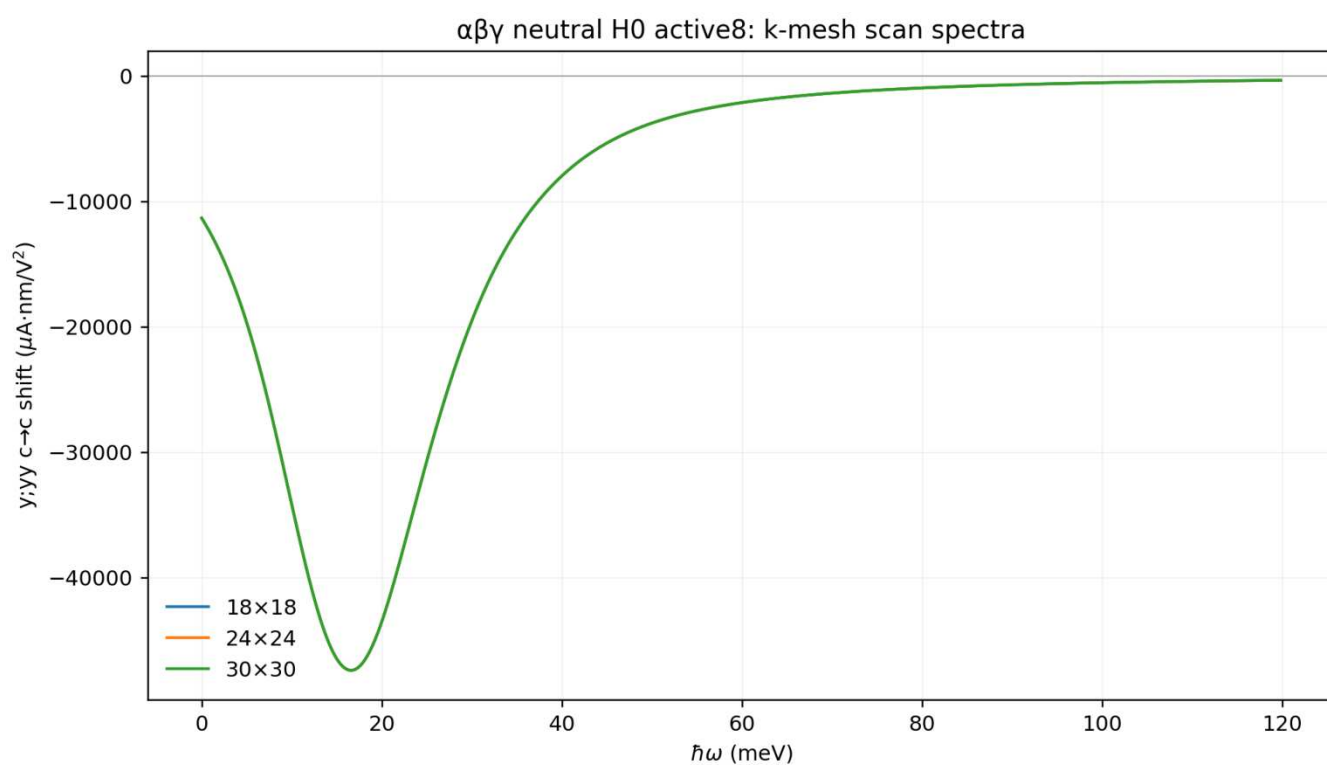


图 3 | $\alpha\beta\gamma$ $\eta=8$ k-mesh spectra。

Historical Figure 2-9 response scale: signed -> -old/2; norms/absolute weights -> old/2

巨大响应是 signed tensor，不是单一正数

- 多个 symmetry-related components 同时很大并具有相反符号。
- Central $c \rightarrow c$ panel 重现低能 total: 不是 remote-band artifact.
- 实验必须指定 polarization、current axis 与 structural domain.

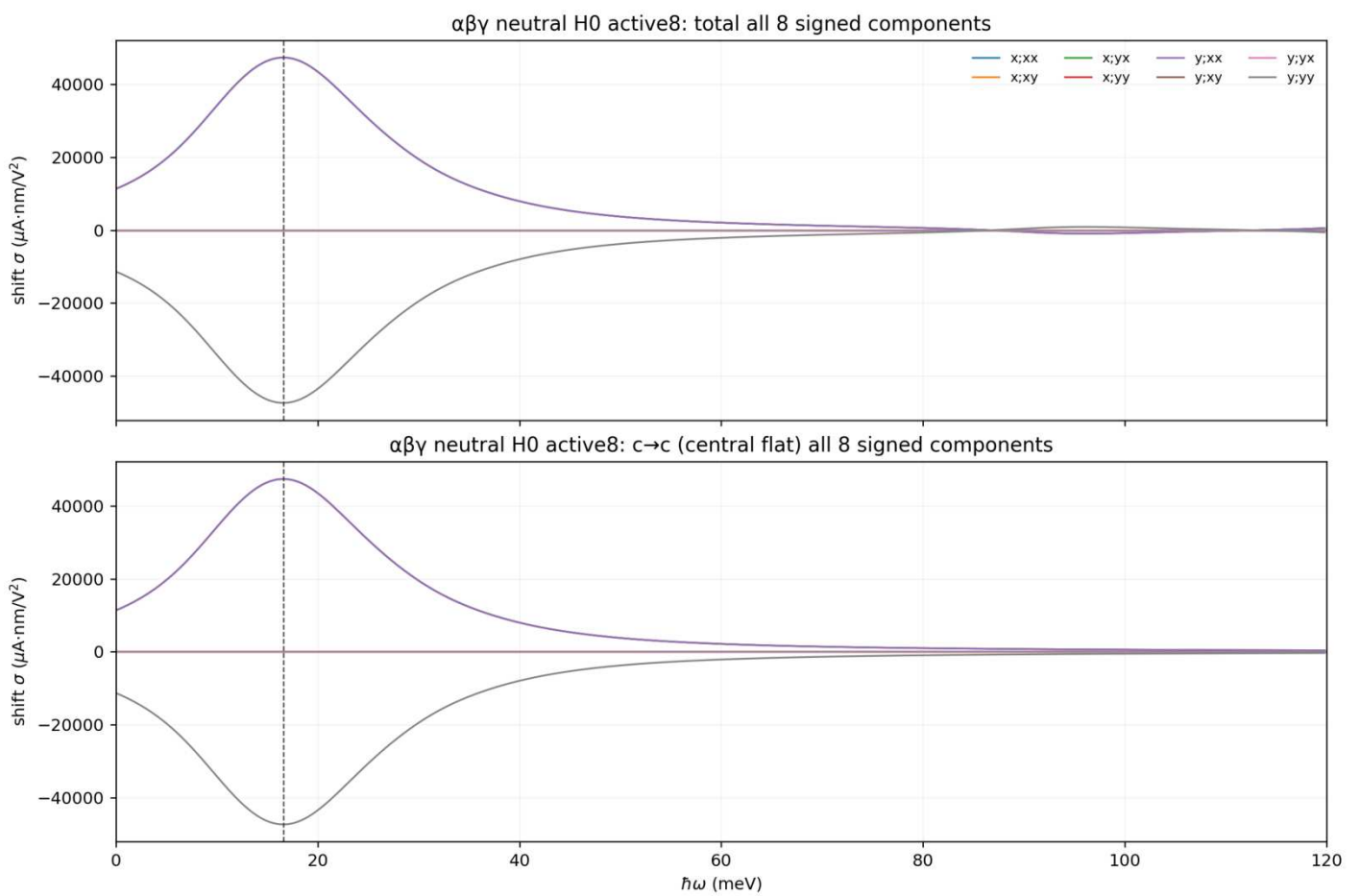


图 4 | Type-II $\alpha\beta\gamma$ 的八个 signed components.

Historical Figure 2-9 response scale: signed $\rightarrow$ -old/2; norms/absolute weights $\rightarrow$ old/2

Mechanism : central-flat resonance , 而非 remote artifact

- JDOS low-energy peak 与 dominant shift component 都由 central $c \rightarrow c$ channel 主导。
- Giant amplitude 还需要 optical weight、covariant shift/geometric factor 与有限 cancellation。

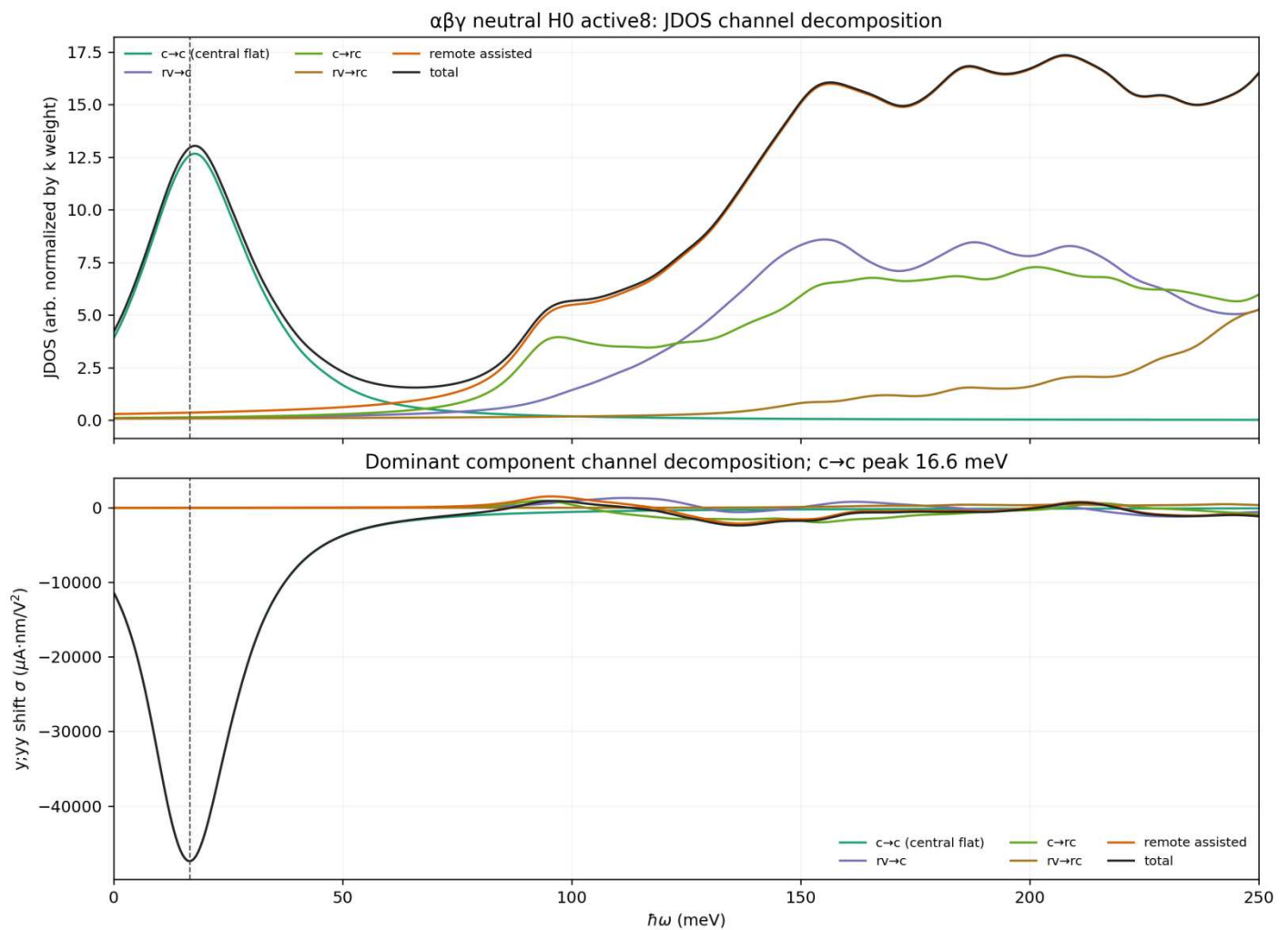
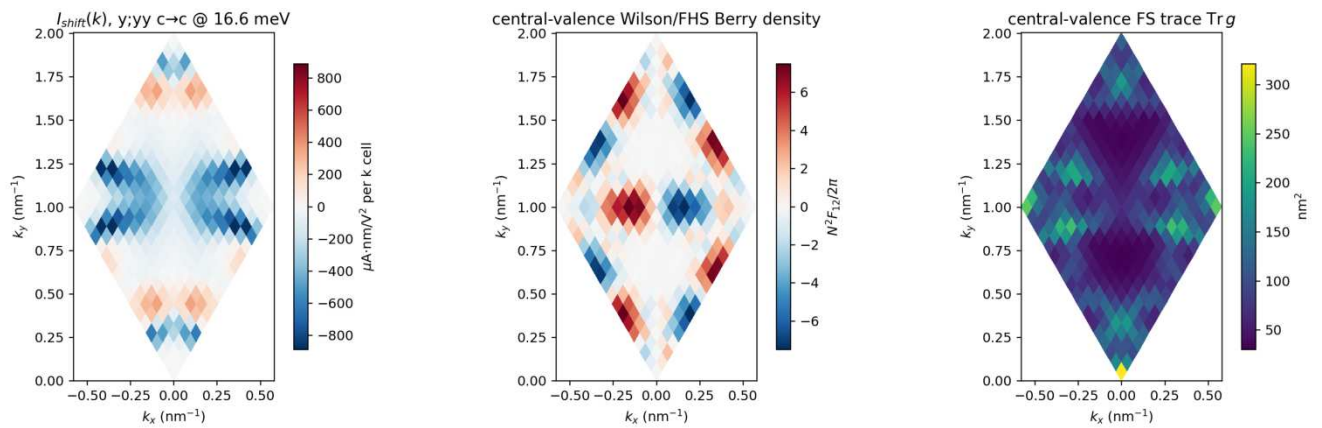


图 5 | JDOS and signed-channel decomposition.

Historical Figure 2-9 response scale: signed -> -old/2; norms/absolute weights -> old/2

Local quantum geometry 是共同底层，不是 Chern 定标

- Berry/FS maps 是 occupied-bundle diagnostics, 不是 pair-resolved transition geometry.
- FS trace 是 finite-difference subspace-fidelity diagnostic; co-location 不是 causal proof.
- 现有 evidence 只说明 Chern alone 不决定 BPVE magnitude.



Chern = $(1/2\pi) \oint_{\text{BZ}} \Omega(k) dk$ 是 unweighted global integral ;

BPVE 是 frequency-resolved、transition-weighted、tensorial signed integral.

图 6 | Shift-current hotspot、Berry density 与 FS trace。

Historical Figure 2-9 response scale: signed $\rightarrow$ -old/2; norms/absolute weights $\rightarrow$ old/2

不加相互作用的 peak 更大，但 HF 主要重排 spectral weight

- Neutral Type-II H0 $\eta=8$ peak $2.3678e4$ ，是 $\epsilon=10$ $v=0$ HF peak $8.9e3$ 的约 2.66 倍。
- HF/H0 integrated low-energy-weight ratio = 0.905：相互作用主要 blue-shift/redistribute。
- Order 仍会改变 frequency、sign、polarization 与 $c \rightarrow c$ / $c \rightarrow rc$ channel content.

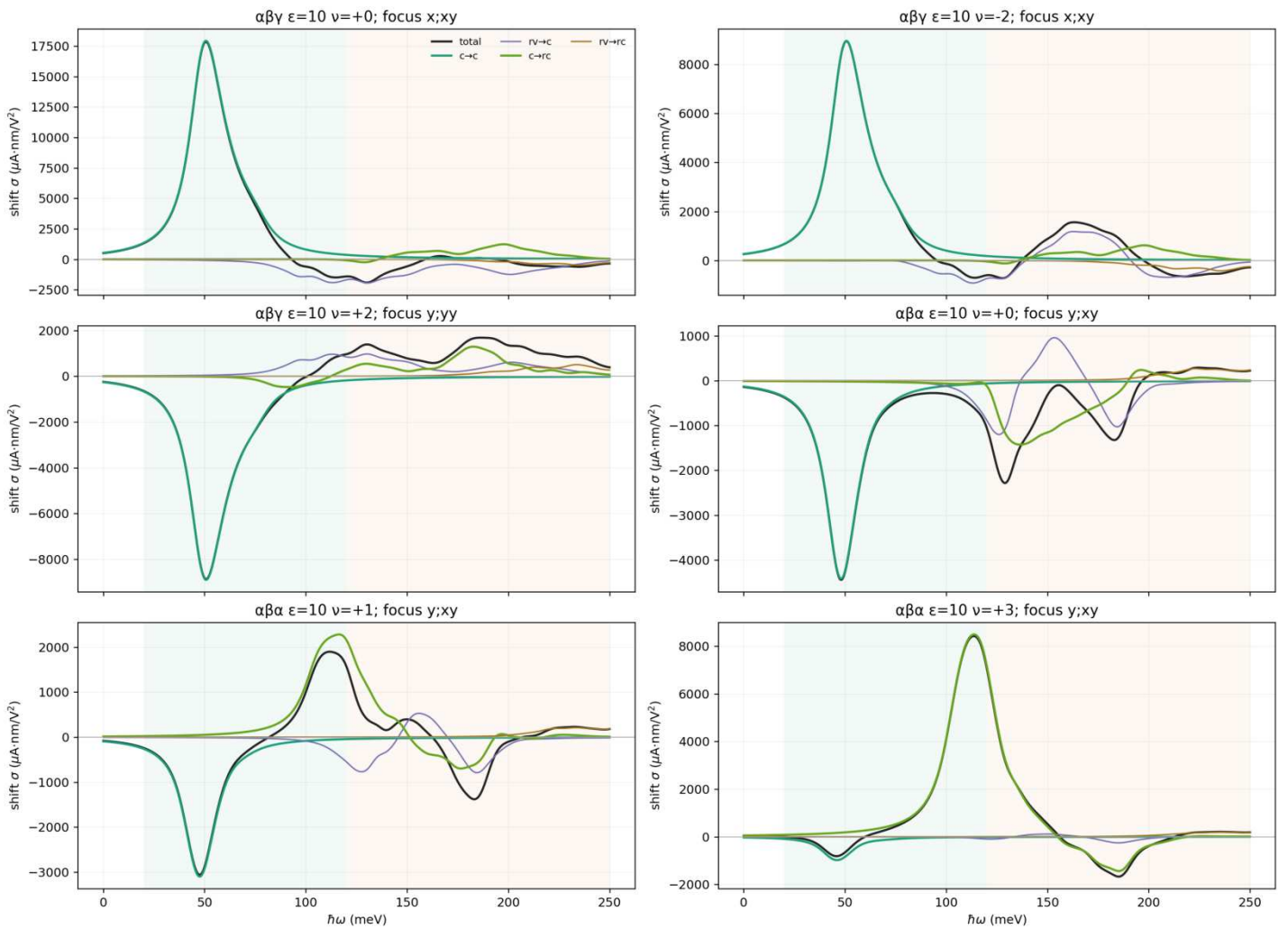


图 7 | active4 $\epsilon=10$ representative HF states.

Historical Figure 2-9 response scale: signed -> -old/2; norms/absolute weights -> old/2

Topology/order 相关，但 brightness 连续变化

- Type-II shown rows 有 active2/active4 Chern agreement: 这是 active-space consistency.
- 它不是 thermodynamic stability; Type-I 集中了 Chern mismatches 与 IVC/KIVC order.
- 不同 Chern labels 不对应量子化 brightness steps.

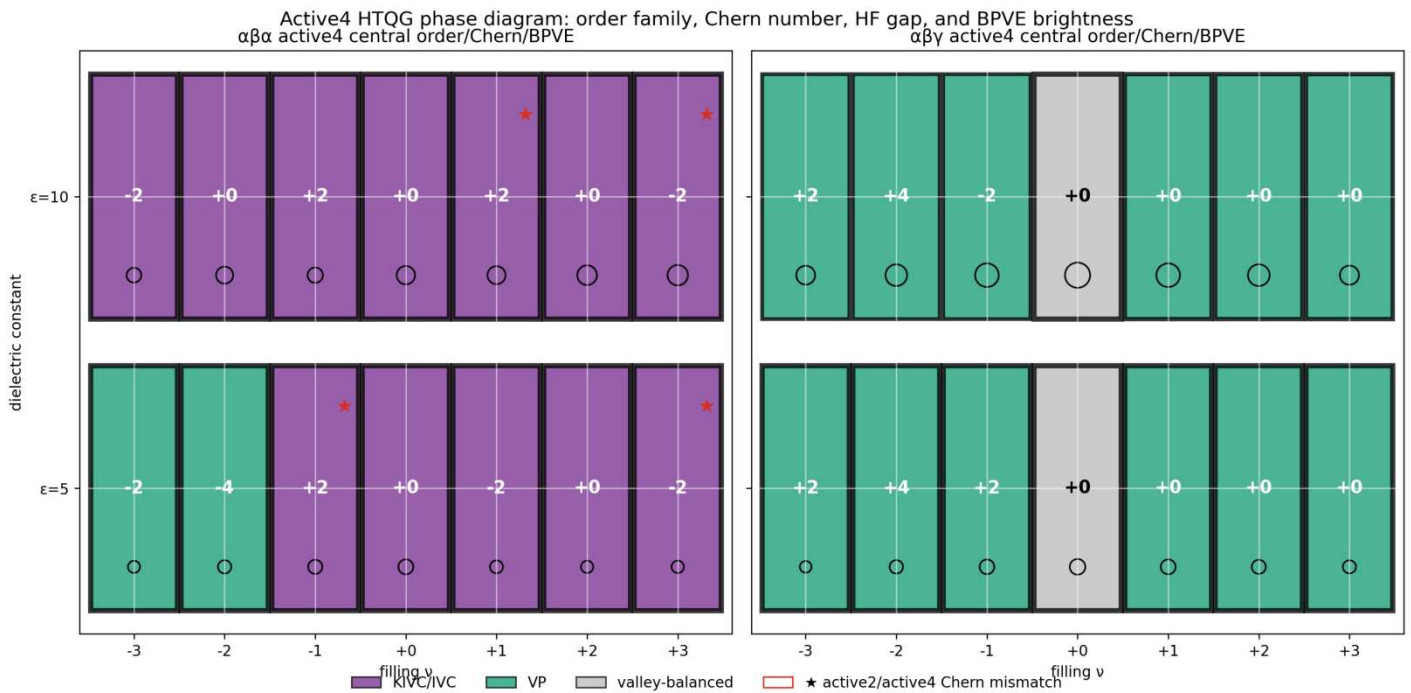


图 8 | Order family、Chern、gap 与 low-energy BPVE weight.

Historical Figure 2-9 response scale: signed $\rightarrow$ -old/2; norms/absolute weights $\rightarrow$ old/2

减弱相互作用可增强 low-energy weight，但需保留 branch caveat

- Along sampled continuations, increasing dielectric ϵ generally increases W_{low} and softens peaks.
- $\epsilon=8/12/15$ seeded from $\epsilon=10$; branch crossings 和 global ground state 未测试。
- 因此这是“弱相互作用有利”的支持 evidence，不是完整 phase-diagram theorem。

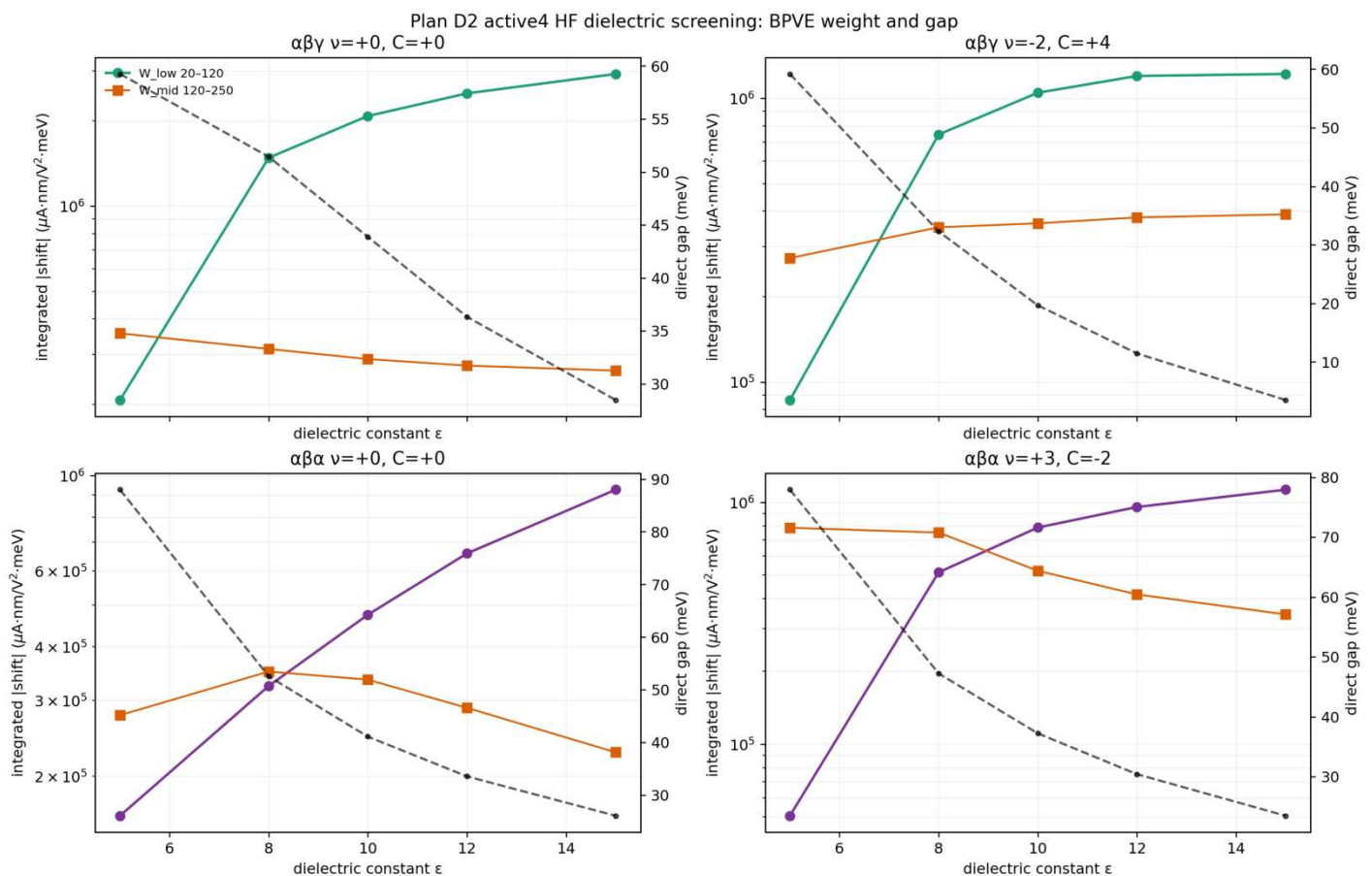


图 9 | Dielectric continuation scan.

Historical Figure 2-9 response scale: signed $\rightarrow$ -old/2; norms/absolute weights $\rightarrow$ old/2

单畴巨大，partner domains 等面积时几乎抵消

- Current-authority Type-II partner tensors 近乎 equal-and-opposite。
- Type-I $\omega < 5$ meV 已排除；剩余曲线仅作 gapless symmetry diagnostic。
- 增大 macroscopic current 的首要条件之一是 single-domain 或 strong domain imbalance。

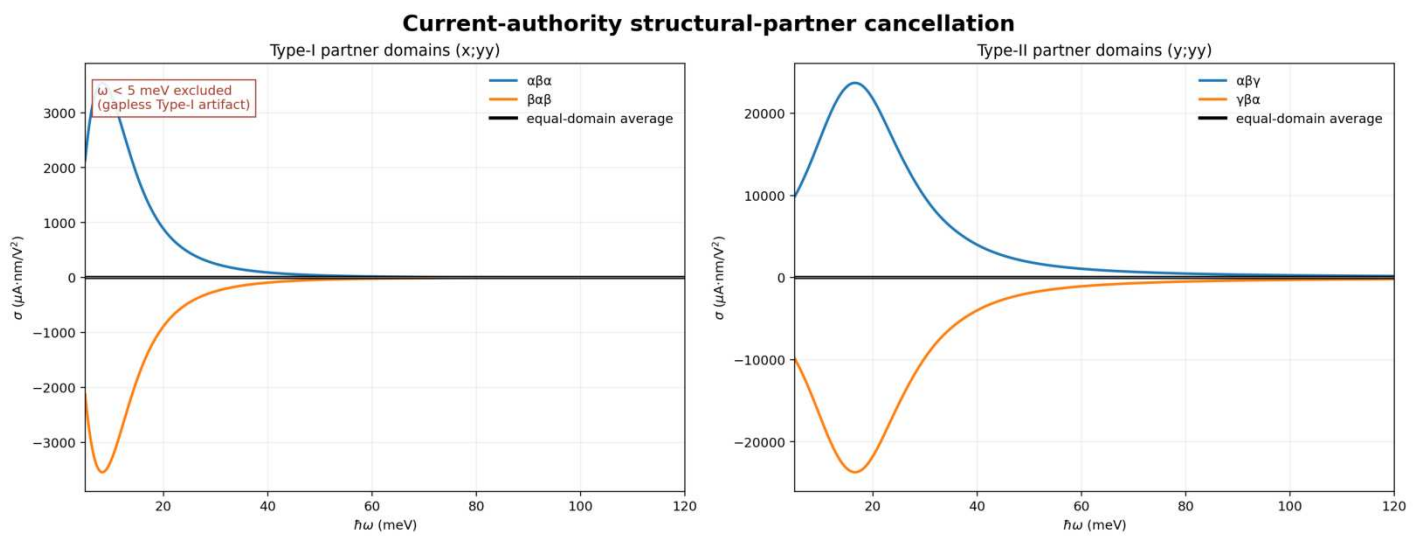


图 10 | Current-authority partner-domain cancellation; Type-I low-frequency exclusion.

只保留 clean Type-II polarization response

- Former Type-I gapless row 已删除，不再把错误 low-frequency coefficient 放入实验图。
- 选择 dominant polarization contraction 可最大化 single-domain current。
- Partner domain 反转 current vector; domain mapping 是实验判据。

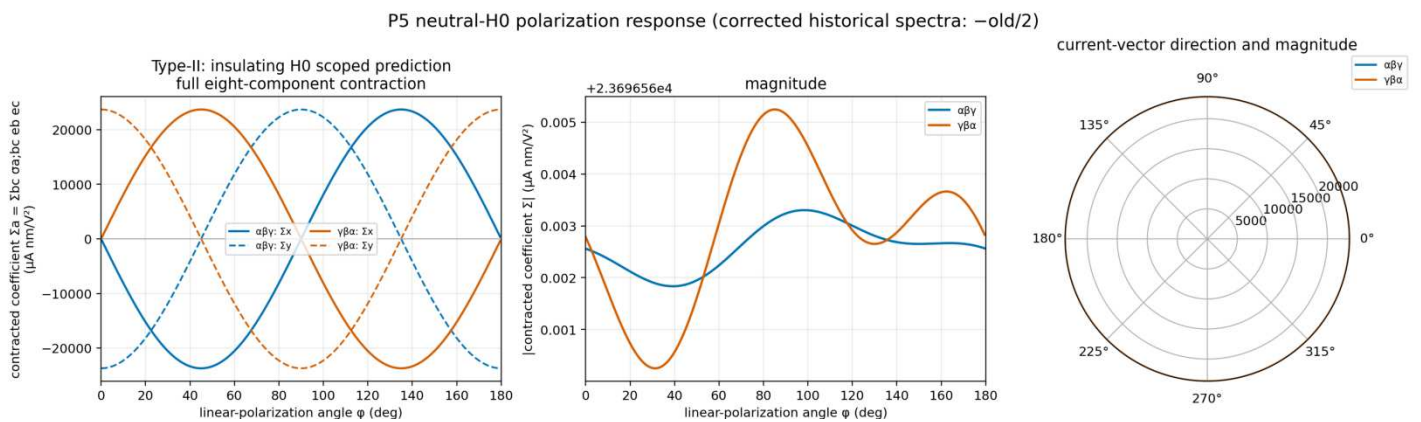


图 11 | Corrected Type-II full-tensor polarization contraction。

高量级结果没有丢失：component、norm、Wlow 全部列出

- $\eta=1$: component 6.876e4; tensor norm 1.375e5 $\mu\text{A}\cdot\text{nm}/\text{V}^2$.
- $\eta=1$: all-eight central- $c \rightarrow c$ Wlow 2.824e6 $\mu\text{A}\cdot\text{nm}\cdot\text{meV}/\text{V}^2$; 不是 peak coefficient.
- Engineering σ/t conversion applies to component or norm; ideal current only uses component.

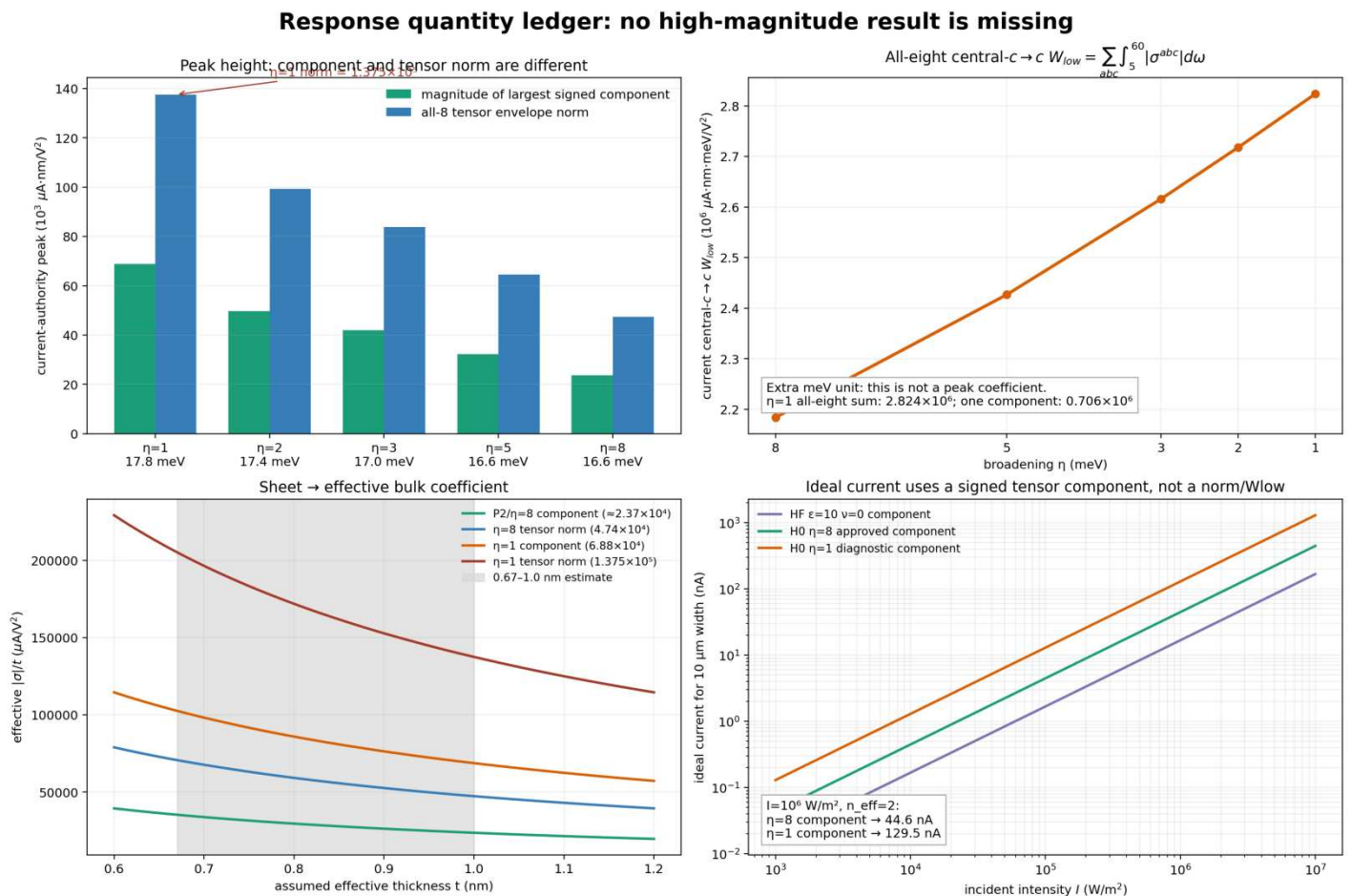


图 12 | Current-authority response ledger directly reconstructed from saved B2 spectra.

如何增大电流，以及怎样与 bulk materials 比较

Increase the measurable component

- Choose Type-II $\alpha\beta\gamma$ near neutrality and avoid the invalid Type-I zero-frequency sector.
- Weaken interaction reshaping through screening; H0 $\eta=8$ peak is $2.66\times$ the shown HF neutral peak.
- Reduce disorder/scattering; smaller η sharpens the peak while W_{low} changes much less.
- Align polarization and select one structural domain; equal partner domains cancel.

Ideal current: component only

- $j_{2D} = \sigma_{component} \times 10^{-15} \times I / (c \epsilon_0 n_{eff})$, $I_{collected} = j_{2D} W$.
- At $I=1e6$ W/m², $n_{eff}=2$, $W=10$ μm : approved $\eta=8 \rightarrow 44.6$ nA; $\eta=1$ diagnostic $\rightarrow 129.5$ nA.
- Do not insert tensor norm or W_{low} into this formula.

Effective bulk comparison, $\sigma_{3D}=\sigma_{2D}/t_{eff}$

P2-approved $\eta=8$ component: $2.37e4$ - $3.53e4$ $\mu\text{A}/\text{V}^2$

$\eta=8$ tensor norm: $4.74e4$ - $7.07e4$ $\mu\text{A}/\text{V}^2$

$\eta=1$ component: $6.88e4$ - $1.03e5$ $\mu\text{A}/\text{V}^2$

$\eta=1$ tensor norm: $1.38e5$ - $2.05e5$ $\mu\text{A}/\text{V}^2$

$t_{eff}=0.67$ - 1.0 nm is a comparison convention. W_{low} is excluded from this table because it retains an extra meV unit.

重写后的核心叙事与下一步

Validated story

- Structural selectivity: Type-II $\alpha\beta\gamma$ finite-frequency BPVE 明显强于 Type-I $\alpha\beta\alpha$ 。
- Legitimacy: finite gaps/frequency + broadening + mesh + active-window checks。
- Mechanism: central $c \rightarrow c$ resonance $\times$ optical/geometric factor $\times$ signed cancellation。
- Interaction: H0 peak 更高; HF shifts/redistributes weight; screening can recover low-energy brightness。
- Experiment: single-domain、polarization-resolved THz current; 避免 Type-I $\omega \approx 0$ artifact。

Needed causal checks

- Matched Type-I/II factorization: JDOS, $|r|^2$, covariant shift term, cancellation。
- Same-order/different-C and same-C/different-order HF comparisons。
- Device-specific absorption, local field, contacts, collection, relaxation, saturation。
- Literature comparison using the same σ definition, broadening, thickness, and polarization convention。

Active6 Type-I P1-P5 remains a controlled supplement, not the giant-BPVE headline.

边界、provenance 与 references

- Type-I $\hbar\omega < 5$ meV remains excluded because the neutral H0 source is gapless.
- $\eta=1/2/3$ peak heights are broadening diagnostics; approved headline is $\eta=8$ P2 component.
- Tensor norm is not one signed component; Wlow must state channel/component scope and has an extra meV.
- Historical/current conversion is quantity-specific: signed -old/2; norm/absolute weight old/2.
- 0.67-1.0 nm thickness is conventional; ideal current omits device effects.
- Only $\alpha\beta\gamma$ is directly P2 full-continuum approved; $\gamma\beta\alpha$ is projected-active8 partner.

Reconciled old high-value figures

B2 $\eta=1$: component/norm/all-eight central-c $\rightarrow$ c Wlow = 6.876e4 / 1.375e5 / 2.824e6.
 Old B2 heatmap was component-resolved: one-component current Wlow max = 0.706e6.
 D1 clean $\alpha\beta\gamma$: central-c $\rightarrow$ c peak 2.498e4; central2_Wlow 2.331e6 with extra meV.
 Active4/D1 Type-I 1e5-1e8 peaks are gapless artifacts.

Figure provenance

Figures 2-9: historical Plan-B products. Figures 1/10/12: current-authority re-plots.
 Figure 11: corrected P5 Type-II crop. Figure 12 is directly derived from saved B2 spectra.

References: Sipe & Shkrebtii (2000); Morimoto & Nagaosa (2016); Chaudhary et al. (2022).

Package: narrative + 12 PNG figures only. No NPZ/JSON/CSV/checkpoints/HF density.